Chapter 2 Study Notes
The Chemical Level of Organization

LEARNING OBJECTIVES

- Introduce the language and fundamental concepts of chemistry
- Discuss how matter is organized
- Discuss how chemical bonds form and how chemical reactions occur
- Compare and contrast organic and inorganic compounds

1. Define matter, chemistry and mass.

2. What are the three forms of matter?

3. Define the term element.

4. What are the essential elements of life (there are 6!). Which of these 6 are the 4 elements that make up the majority of our body.

5. Provide some examples of trace elements and why they are essential.

6. What is the core structure of atoms?

7. Define:
    a. Proton
    b. Neutron
    c. Electron

8. Define:
    a. Atomic number
    b. Atomic Mass
    c. Isotope

9. What is an ion? Compare and contrast cation vs anion.

10. What is a free radical?

11. Define and give examples of radioisotopes.

12. What is a valence shell? Why is it important?

13. How many electrons are in the valence shell of Magnesium? Boron? Potassium, phosphorus?

14. Draw a Bohr model for each of the above atoms

15. Define ionic bond, covalent bond, polar covalent, nonpolar covalent, and hydrogen bonds. Give examples of each of these bond types.

16. Hydrogen bonds give water special properties. What properties do water molecules have due to the presence of hydrogen bonds.

17. In a chemical reactions, there are reactants and products. Research a chemical reaction (general) and label the reactants and the products.

18. What is metabolism? What is anabolism and catabolism?

19. List the 4 types of chemical reactions discussed in class. Define them. Provide an example of each type of chemical reaction.

20. Define energy. What are the different types of energy found in nature?

21. What is meant by an exergonic reaction? Endergonic reaction? Sketch a graph for each of these reaction types (Time on the X axis and potential energy, ΔG on the y axis).

22. What is the activation energy? How do enzymes affect this energy type?

23. What is an enzyme? What are some common core characteristics of enzymes?

24. List the steps of a general enzymatic reaction.

25. What properties of water make it considered to be the universal solvent.

26. Define acid and base.

27. What is a pH scale?

28. If a solution has a high pH, is it an acid or base? Low pH? Compare and contrast the [H+] in an acid vs a base solution.

29. What is a buffer? Why are buffers key to all living organisms? Give an example of use of a buffer system in the human body.

30. Create a chart of the most common functional groups found on the 4 major macromolecules. Give examples where each are found.

31. Create a chart of the macromolecules and the monomers that make them up. Also include the special bonds that are found linking them.

32. What is the role of carbohydrates in the human body? What are key characteristics of this macromolecule?

33. What is the role of Lipids in the human body? What are key characteristics of this macromolecule?

34. What is the role of proteins in the human body? What are key characteristics of this macromolecule?

35. What are some of the roles of proteins in the human body?

36. What is the role of nucleic in the human body? What are key characteristics of this macromolecule?

37. How many amino acids are found in nature? Draw the common core structure of an amino acid. Label the amino group, central carbon, functional group and carboxyl group.

38. How many nucleotides are found in the human body? Draw the common structure of a nucleic acid found in DNA. RNA? Label the nitrogenous base, ribose/deoxyribose sugar, and phosphate. Draw a circle around the major difference of the sugar base in a DNA vs RNA molecule.

39. Draw a molecule of ATP. Why is this molecule so central to life. What are the ways we can make ATP that were discussed in class?s